Data Science
Fundamentals

Introduction to Data and Random Variables

The foundation of information in AI and Machine Learning



Made with Napkin

Classifying Data by Nature







Qualitative Data

Describes qualities and categories

Quantitative Data

Expresses information in numbers

Quantitative data is further classified as discrete, which includes countable values, or continuous, which can take any value within a range.

Made with Napkin

Time-Based Data



Cross-sectional Data

Data collected at one particular point in time from different individuals or objects, such as student heights in 2026.

Time Series Data

Data collected for the same variable over different points in time, like tracking a company's yearly sales growth.

Panel Data

A combination of both, where different subjects are tracked over different points in time, such as the sales of three companies over four years.

Made with Napkin

Understanding Random Variables

Discrete Random Variable

A variable that can take countable values, such as the number of heads when tossing three coins.

Continuous Random Variable

A variable that can take any value within a range, such as the height of a student or the temperature of a room.

Representing uncertainty through outcomes

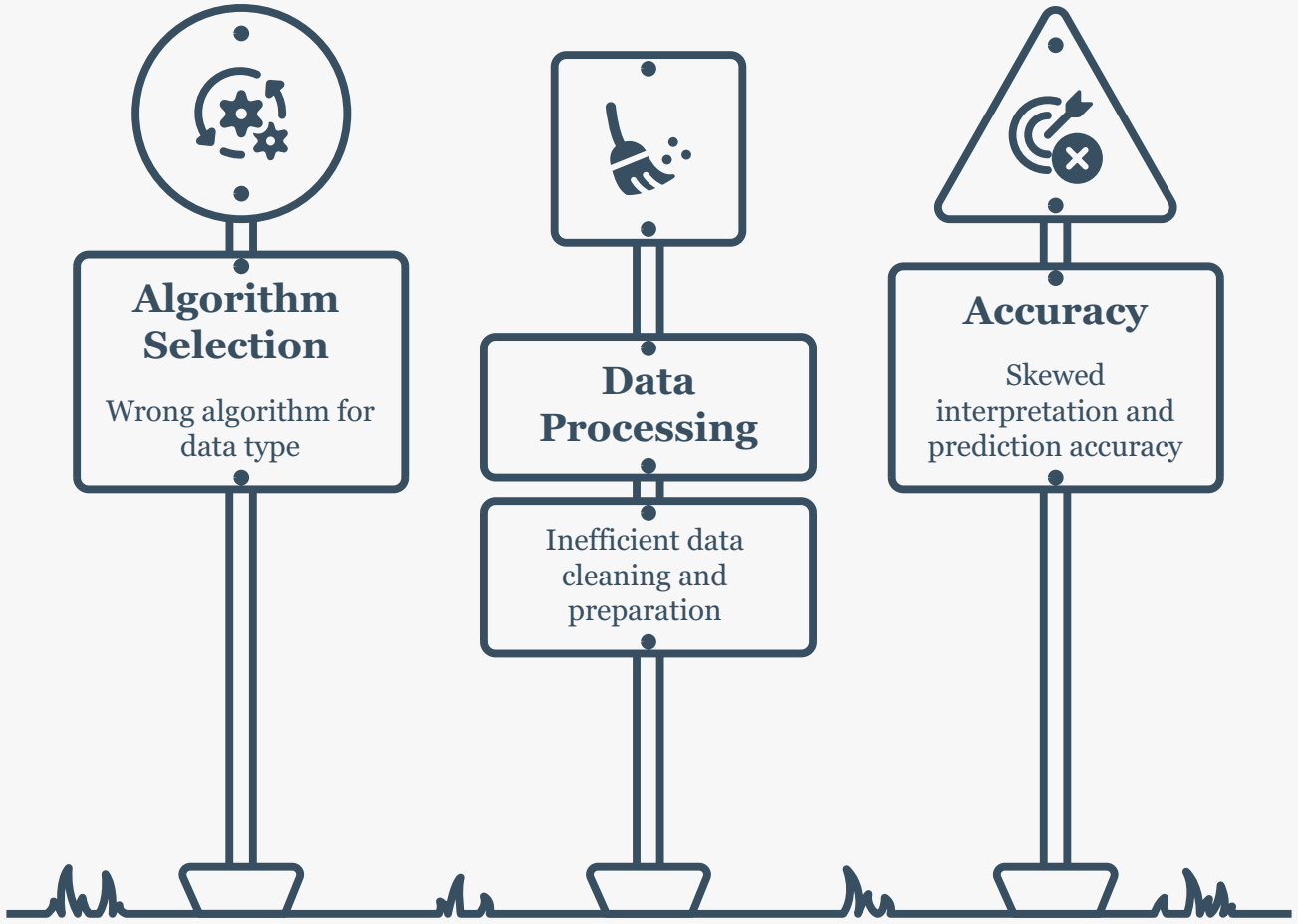
A random variable is a variable whose value is determined by the outcome of a random experiment. We use them to represent events where the outcome is not known beforehand.

Made with Napkin

Why Data Types Matter in AI/ML

Choosing the right tools for the job

The type and structure of your data are the primary factors in determining how an AI model should be built and processed.



Made with Napkin

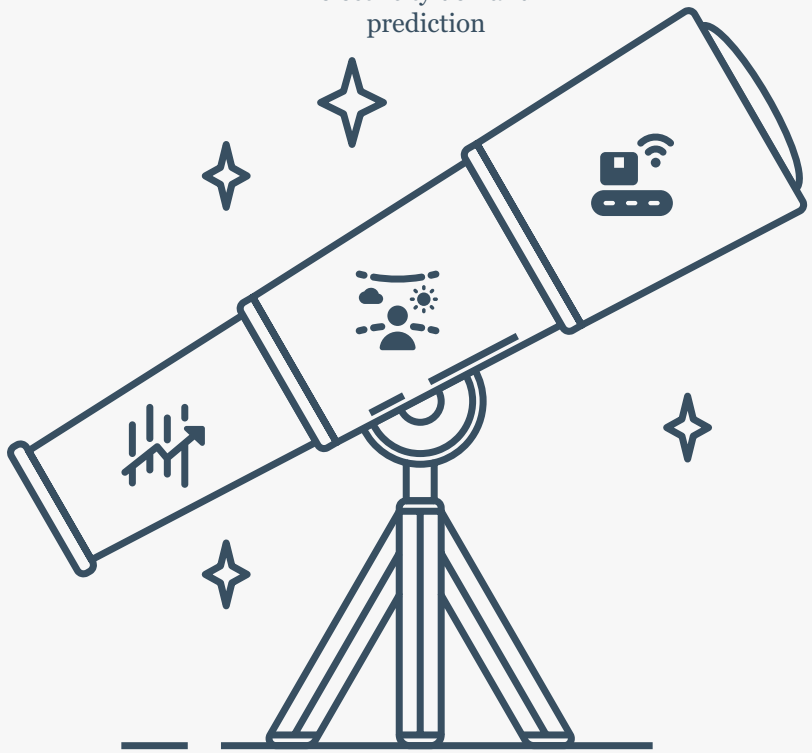
The Power of Time Series in AI

Environmental Science

Weather forecasting and electricity demand prediction

Financial Markets

Predicting stock prices and market trends



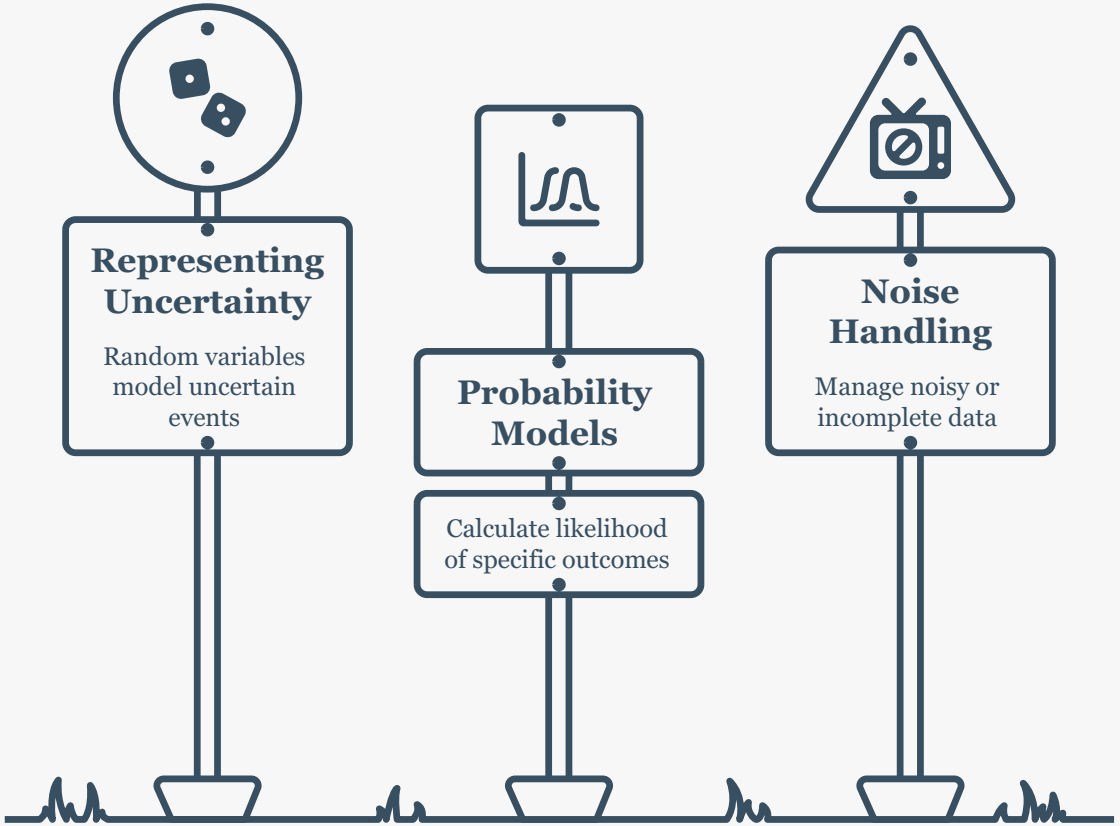
Industrial AI

Predictive maintenance using sensor readings

In these models, the order of observations is critical.

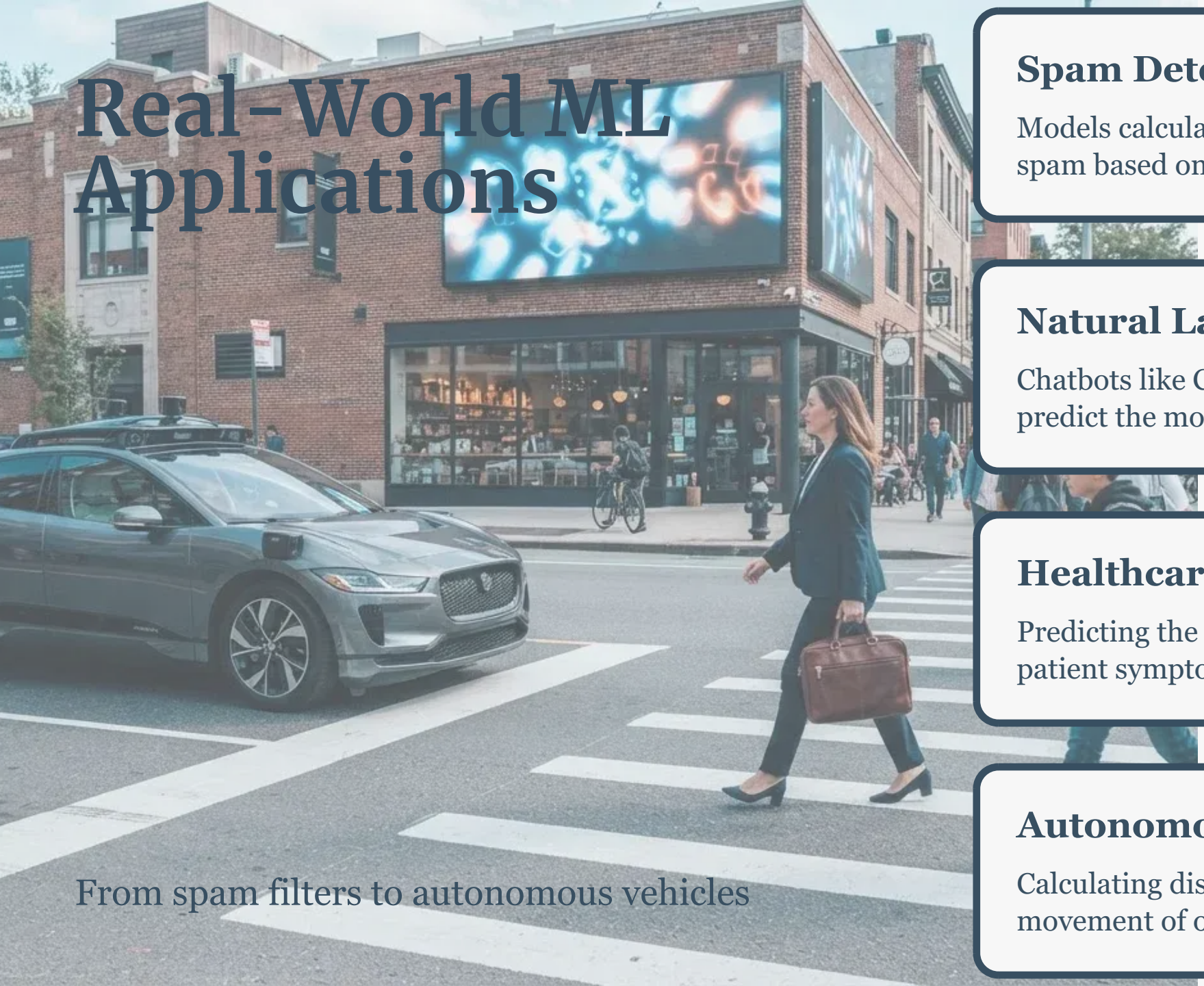
Made with Napkin

How to handle uncertainty and noise in ML models?



Made with Napkin

Real-World ML Applications



From spam filters to autonomous vehicles

Spam Detection

Models calculate the probability that an email is spam based on its content.

Natural Language Processing

Chatbots like ChatGPT use random variables to predict the most likely next word.

Healthcare

Predicting the likelihood of a disease based on patient symptoms and history.

Autonomous Vehicles

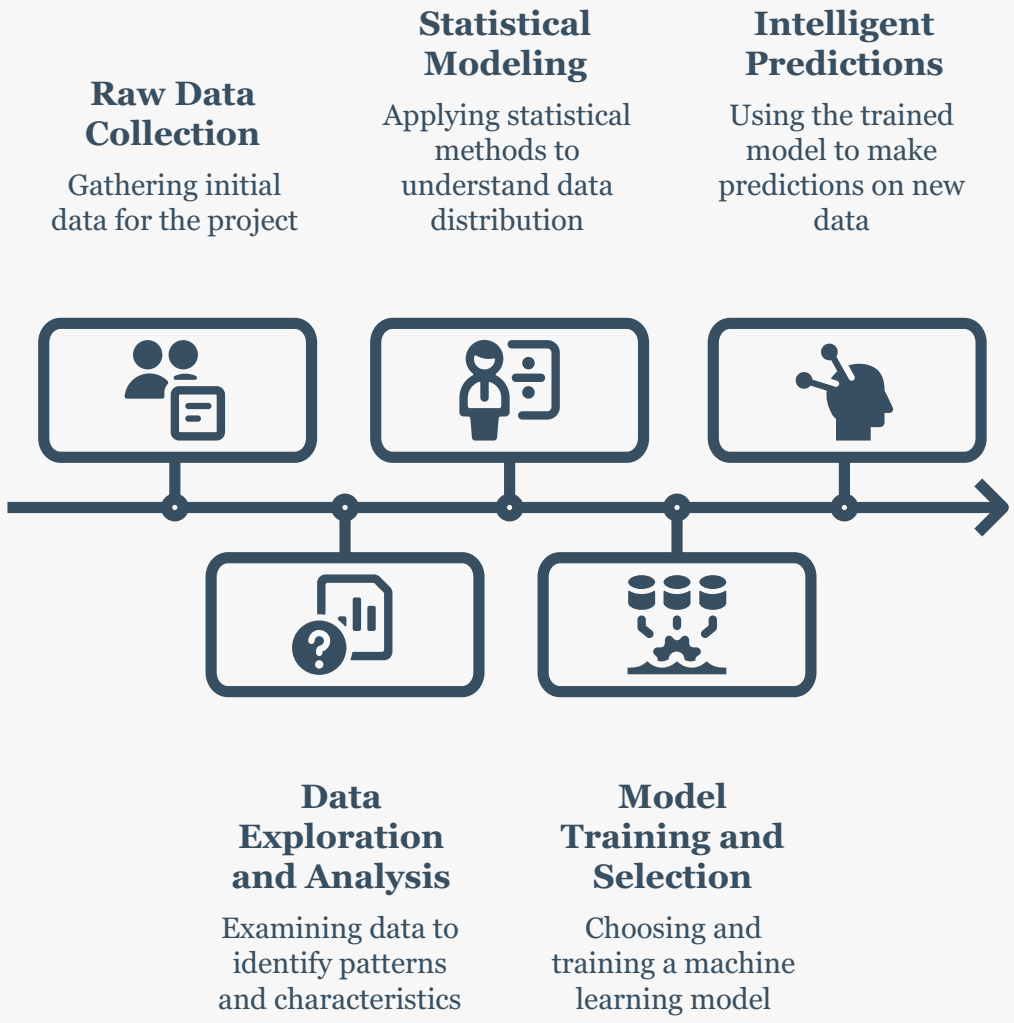
Calculating distances and predicting the movement of other objects on the road.

Made with Napkin

Summary: The AI/ML Pipeline

From raw data to intelligent predictions

Understanding these fundamental concepts is the first step in any successful AI or Machine Learning project. By correctly identifying your data type and its distribution, you can choose the most effective ML technique to solve your specific problem.



Made with Napkin